

Simplex method

Main idea: Go from one basis to another basis, trying to increase objective function.

Start: Feasible basis

Each iteration: ① Find an entering variable: one with positive coefficient in the objective. (Possible to raise objective value.)

② Find a leaving variable: calculate the ratios between the RHS and the positive entries in the column of the entering variable. A basic variable in a row with the minimum ratio is the leaving variable.

③ Find the canonical form with respect to the new basis from the original LP. Repeat.

$$\begin{aligned} \max \quad & (3 \quad 4 \quad -1 \quad 0 \quad 0)x \\ \text{s.t.} \quad & \begin{pmatrix} 4 & 7 & 4 & 1 & 0 \\ 2 & -3 & 5 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 8 \\ 6 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Basis $\{4, 5\}$ BFS $(0, 0, 0, 8, 6)^T$

□ Pick x_1 as entering var.

$$\min \left\{ \frac{8}{4}, \frac{6}{2} \right\} = 2. \leftarrow$$

□ Pick x_4 as leaving var.

$$\begin{aligned} [x_1 = t, x_4 = 8 - 4t, x_5 = 6 - 2t. \\ \Rightarrow t \leq \frac{8}{4} \quad t \leq \frac{6}{2} \Rightarrow t = 2] \end{aligned}$$

New basis $\{1, 5\}$

Terminating conditions for simplex:

① If all coefficients in the objective function are non-positive, then we have an optimal solution.

$$\begin{aligned} \max \quad & (0 \quad -1 \quad 0 \quad -3)x + 7 \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 3 & 0 & 1 \\ 0 & 4 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 5 \\ 9 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

Basis $\{1, 3\}$. BFS $(5, 0, 9, 0)^T$. Obj value 7.

Any feasible x has obj value

$$(0 \quad -1 \quad 0 \quad -3)x + 7 \leq 7. \Rightarrow \text{BFS optimal.}$$

$\leq 0^T \quad \geq 0$

② If the column in A of the entering variable is non-positive, the LP is unbounded.

$$\begin{aligned} \max \quad & (0 \quad 3 \quad 0 \quad 2)x + 5 \\ \text{s.t.} \quad & \begin{pmatrix} 1 & 0 & 0 & 7 \\ 0 & -4 & 1 & 3 \end{pmatrix} x = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \\ & x \geq 0 \end{aligned}$$

x_2 enters. Let $x_2 = t$.

$$\begin{aligned} \Rightarrow \quad & x_1 = 1 \\ & x_2 = t \\ & x_3 = 2 + 4t \\ & x_4 = 0 \end{aligned}$$

No limit on t .

$$x = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 4 \\ 0 \end{pmatrix}$$

Certificate of unboundedness

Termination of simplex

Does simplex always terminate? Not always. In rare cases, cycling occurs and simplex cannot end.

Problem: Degenerate iterations. BFS does not change in an iteration (basis changes). No improvement on the objective value.

Cycling: basis $B_1 \rightarrow B_2 \rightarrow \dots \rightarrow B_K \rightarrow B_1 \rightarrow \dots \rightarrow B_K \rightarrow B_1 \rightarrow \dots$
degenerate

Solution: Use Bland's Rule when selecting entering / leaving variables. Whenever there is a choice, we pick a variable with the lowest index.

Example: $\max (-2 \ 3 \ 0 \ 1 \ 0 \ 0)x$
s.t. $\begin{pmatrix} 2 & 1 & 1 & 5 & 0 & 0 \\ 3 & 2 & 0 & 6 & 1 & 0 \\ 4 & 3 & 0 & 7 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 10 \\ 4 \\ 6 \end{pmatrix}$
 $x \geq 0$

Two choices of entering vars: x_2, x_4 .

Bland's rule picks x_2 .

$\min \left\{ \frac{10}{1}, \frac{4}{2}, \frac{6}{3} \right\} = 2$. Both x_5, x_6 could leave. Bland's rule picks x_5 .